Rates of Change & Concavity

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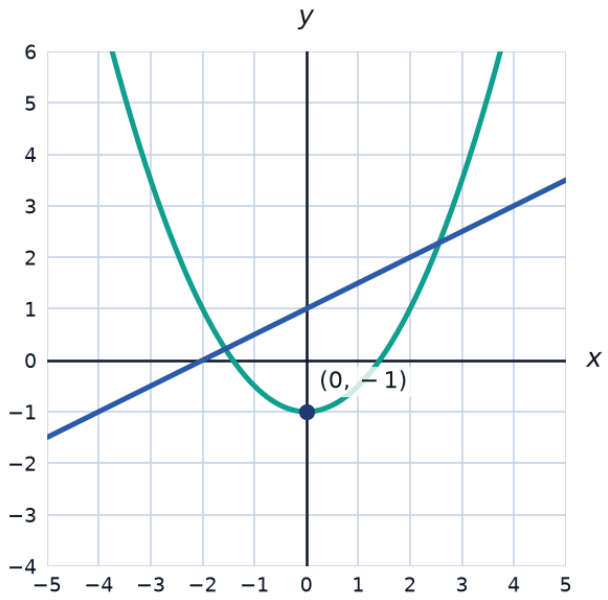
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Average rate of change**

Find the average rate of change of $f(x) = x^2 + 2x$ on the interval $[1, 4]$.

2 **[Foundational]** • **Target: Average rate of change**

A population grows according to $P(t) = 200(1.05)^t$ (years). Find the average rate of change of the population from $t = 0$ to $t = 10$, to the nearest whole number.



3 **[Foundational]** • **Target: Increasing/decreasing behavior from rate of change**

A function f has average rates of change of 2 on $[0, 1]$, 5 on $[1, 2]$, and 9 on $[2, 3]$. Determine whether f appears to be concave up or concave down on $[0, 3]$, and explain.

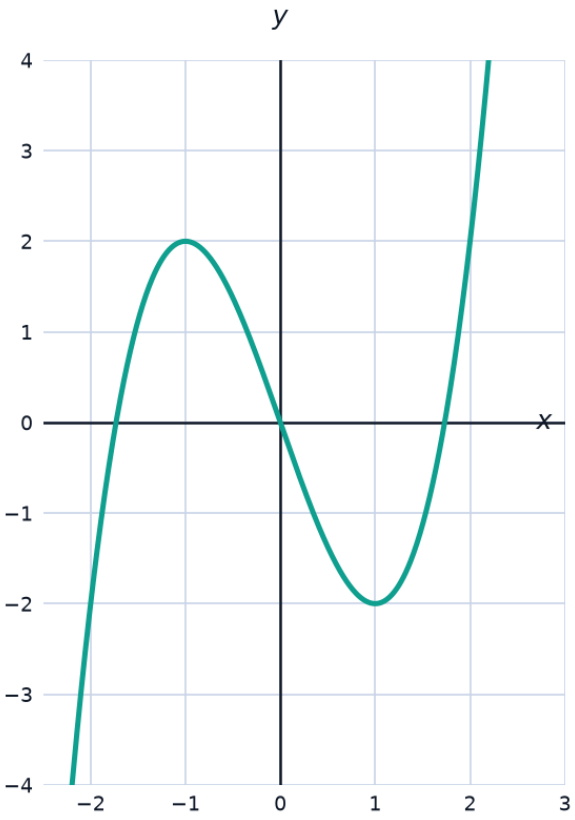
4 **[Foundational]** • **Target: Increasing/decreasing behavior from rate of change**

A different function g has average rates of change of 8 on $[0, 1]$, 5 on $[1, 2]$, and 1 on $[2, 3]$. Classify the concavity of g on $[0, 3]$.

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5 [Foundational] • Target: Visualizing concavity and inflection

The graph shows $f(x) = x^3 - 3x$, which changes concavity at $x = 0$.



a Describe the concavity of the graph on $(-2, 0)$.

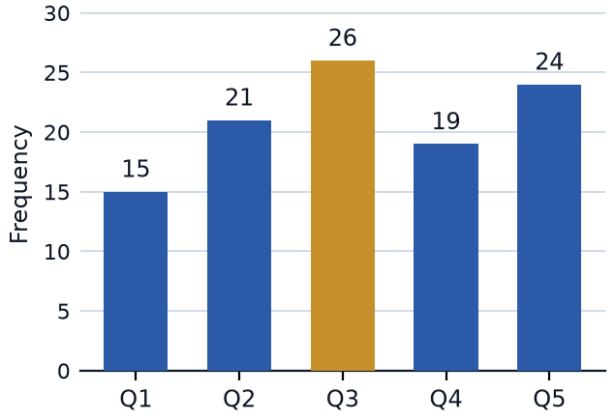
b Describe the concavity of the graph on $(0, 2)$.

c State what kind of point $x = 0$ represents.

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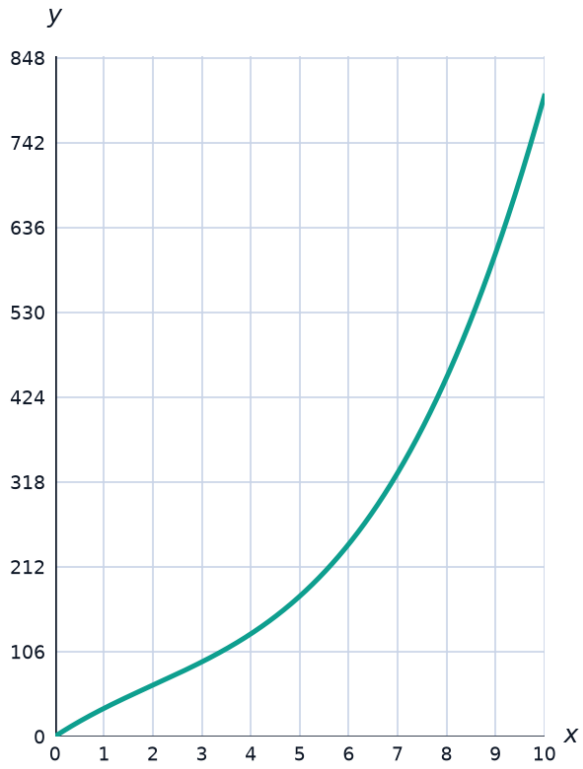
6 [Foundational] • Target: Rate of change of the rate of change

Using average rates of change over consecutive equal-width intervals as a proxy for how the function's growth rate itself is changing, explain what it means for a function to be "concave up" purely in terms of consecutive AROC values, without any calculus notation.



7 [Foundational] • Target: Applying rate-of-change reasoning to models

A company's revenue (in thousands) is modeled by $R(t) = t^3 - 6t^2 + 40t$ for $0 \le t \le 10$ years. Find the average rate of change on $[0, 2]$ and on $[2, 4]$, then determine whether revenue growth is accelerating or decelerating over $[0, 4]$.



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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 **[Intermediate]** • **Target: Applying rate-of-change reasoning to models**

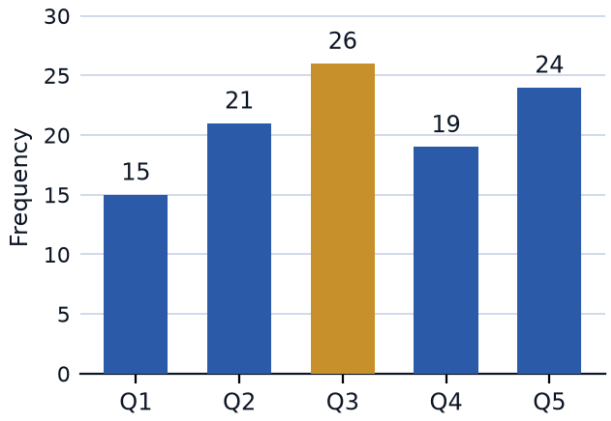
For the same revenue function, find the AROC on $[4, 6]$ and compare it to the AROC on $[2, 4]$ found above, to determine the concavity trend as t increases past 4.

9 **[Intermediate]** • **Target: Estimating rate of change from a table**

A car's odometer reading (miles) is recorded every hour: $t = 0$: 120; $t = 1$: 175; $t = 2$: 238; $t = 3$: 310. Find the AROC over each hour and determine whether the car's speed is increasing, decreasing, or constant.

10 **[Intermediate]** • **Target: Estimating rate of change from a table**

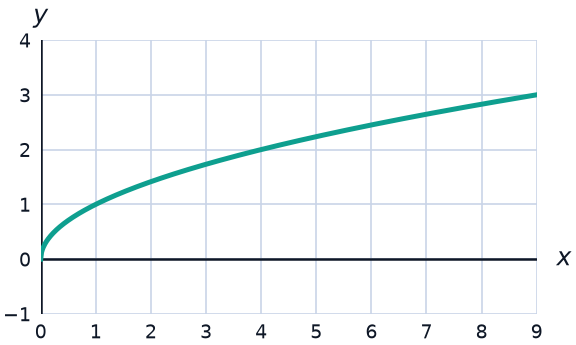
Using the same table, find the AROC over the full interval $[0, 3]$ and explain why it differs from the individual hourly rates.



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11 **[Intermediate]** • **Target: Visualizing rate of change on a graph**

The graph shows $f(x) = \sqrt{x}$, whose average rate of change decreases as x increases.



a Find the AROC of $f(x) = \sqrt{x}$ on $[0, 1]$ and on $[4, 9]$.

b Explain how the shrinking steepness in the graph relates to your two AROC values.

c State whether $f(x) = \sqrt{x}$ is concave up or concave down, based on this pattern.

12 **[Intermediate]** • **Target: Concavity and real-world interpretation**

A company reports that its costs are increasing, but at a decreasing rate. Sketch what this implies about the concavity of the cost function, and explain your reasoning without using calculus notation.

13 **[Intermediate]** • **Target: Average rate of change**

Find the average rate of change of the polar radius $r(\theta) = 3 + 2\sin(2\theta)$ over the interval $[0, \frac{\pi}{4}]$, and interpret whether the distance from the origin is increasing or decreasing on this interval.

14 **[Intermediate]** • **Target: Increasing/decreasing behavior from rate of change**

Solve the exponential equation $9^x - 4 \cdot 3^{x+1} + 27 = 0$ for all real x.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

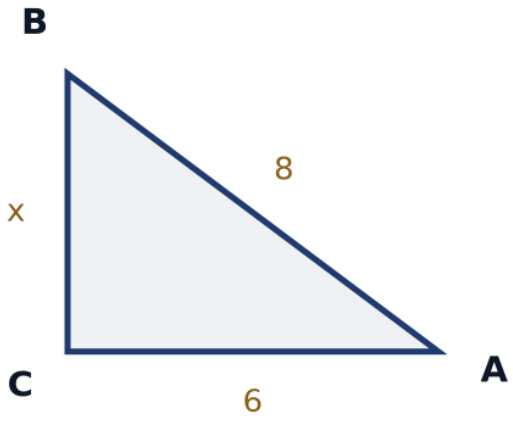
15 **[Challenging]** • **Target: Visualizing concavity and inflection** • **Cross-Lesson: 4.02 Piecewise-Defined & Implicit Functions**

Evaluate $\log_2(5) \cdot \log_5(9) \cdot \log_3(16)$ without a calculator using the change-of-base formula.

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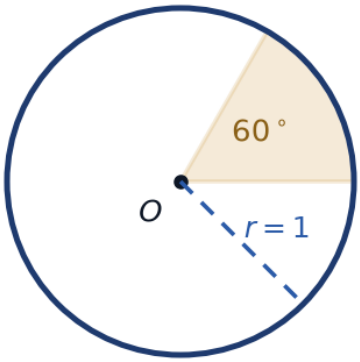
16 [Challenging] • Target: Rate of change of the rate of change • Cross-Lesson: 4.03 Parametric Equations & Modeling

A right-angled triangle ABC is shown with the right angle at vertex C. The length of side AC is 8 units and the length of side BC is 6 units. What is the value of $\sin(\angle BAC)$?



17 [Challenging] • Target: Applying rate-of-change reasoning to models • Cross-Lesson: 1.01 Polynomial End Behavior & Zeros

A point P lies on the unit circle such that its terminal ray makes an angle of θ with the positive x-axis. If the x-coordinate of P is $\frac{1}{2}$, what is the y-coordinate of P? Assume θ is in the first quadrant.



18 [Challenging] • Target: Estimating rate of change from a table • Cross-Lesson: 1.02 Rational Functions & Asymptotes

Find the exact value of $\cos\left(\frac{7\pi}{12}\right)$ and $\sin\left(\frac{7\pi}{12}\right)$ using angle addition identities.

19 [Challenging] • Target: Visualizing rate of change on a graph • Cross-Lesson: 2.01 Exponential Functions & Growth Models

A Ferris wheel has a diameter of 40 meters and its lowest point is 4 meters above the ground. Passengers board at the lowest point at $t = 0$, and the wheel completes one full revolution every 8 minutes. Write a cosine model $h(t)$ for a passenger's height at time t, and find the times during the first revolution when $h(t) = 34$ meters.

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20 [Challenging] • Target: Concavity and real-world interpretation • Cross-Lesson: 2.02 Logarithmic Functions & Properties

Find the exact angle θ between $\mathbf{u} = \langle 3, 1 \rangle$ and $\mathbf{v} = \langle 1, 2 \rangle$, and compute the vector projection $\text{proj}_{\mathbf{v}}\mathbf{u}$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Average rate of change

Find the average rate of change of $f(x) = -7x^2$ on $[-4, -1]$.

a. 35

b. -21

c. 33

d. 37

22 [Foundational] • Target: Increasing/decreasing behavior from rate of change

Find the average rate of change of $f(x) = 9x^2 + 2x$ on $[5, -3]$.

a. 18

b. 20

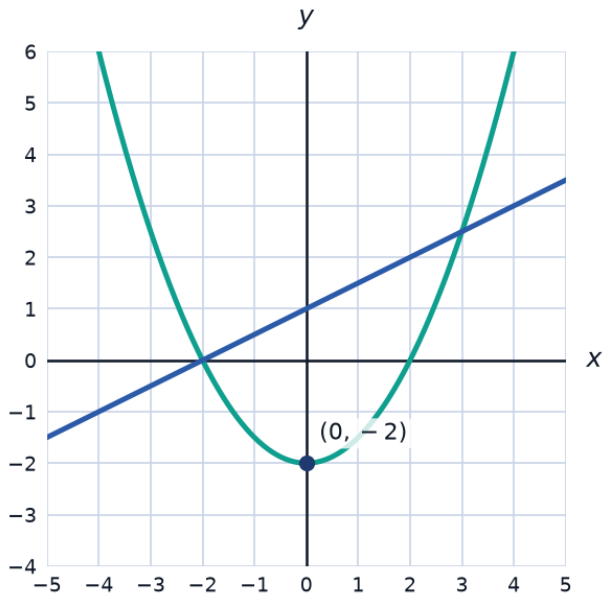
c. -70

d. 22

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23 [Foundational] • Target: Visualizing concavity and inflection

Find the average rate of change of $f(x) = -9x^2 + x$ on $[4, 0]$.



a. -35

b. -33

c. -37

d. 37

24 [Foundational] • Target: Rate of change of the rate of change

Find the average rate of change of $f(x) = 2x^2 - 5x$ on $[6, -3]$.

a. -1

b. -23

c. 1

d. 3

25 [Foundational] • Target: Applying rate-of-change reasoning to models

Find the average rate of change of $f(x) = 6x^2$ on $[3, -3]$.

a. 0

b. -36

c. 2

d. -2

26 [Foundational] • Target: Estimating rate of change from a table

Find the average rate of change of $f(x) = -9x^2 - 9x$ on $[-4, 0]$.

a. -45

b. 25

c. 29

d. 27

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27 [Foundational] • Target: Visualizing rate of change on a graph

Find the average rate of change of $f(x) = x^2 + 4x$ on $[5, 4]$.

a. 11

b. 15

c. 13

d. 3

28 [Foundational] • Target: Concavity and real-world interpretation

Find the average rate of change of $f(x) = -2x^3$ on $[2, -5]$.

a. -38

b. -42

c. -18

d. -34

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Average Rate of Change (AROC)

Find the average rate of change of $f(x) = -x^3$ on $[0, 5]$.

a. -21

b. -25

c. -27

d. -23

30 [Intermediate] • Target: Increasing/Decreasing and Changing Rates

Find the average rate of change of $f(x) = 3x^3$ on $[-2, 3]$.

a. 27

b. 15

c. 21

d. 3

31 [Intermediate] • Target: Concavity & Inflection Points

Find the average rate of change of $f(x) = 2x^3$ on $[-4, 1]$.

a. 30

b. 22

c. 26

d. 18

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32 [Intermediate] • Target: Contextual Rates of Change Applications

Find the average rate of change of $f(x) = -5x^3$ on $[-2, 3]$.

a. -25

b. -5

c. -35

d. -45

33 [Intermediate] • Target: Average rate of change

Find the average rate of change of $f(x) = -5x^3$ on $[0, -3]$.

a. -47

b. -35

c. -55

d. -45

34 [Intermediate] • Target: Increasing/decreasing behavior from rate of change

Find the average rate of change of $f(x) = 6x^3$ on $[3, -5]$.

a. 24

b. 102

c. 126

d. 114

35 [Intermediate] • Target: Visualizing concavity and inflection

A car's odometer reading (miles) is recorded every hour: $t = 0$: 75; $t = 1$: 111; $t = 2$: 138; $t = 3$: 156. Is the car's speed increasing, decreasing, or constant?

a. Decreasing

b. Increasing

c. Constant

d. Cannot be determined from this data

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36 [Intermediate] • Target: Rate of change of the rate of change

A car's odometer reading (miles) is recorded every hour: $t = 0$: 151; $t = 1$: 174; $t = 2$: 197; $t = 3$: 220. Is the car's speed increasing, decreasing, or constant?

a. Increasing

b. Constant

c. Decreasing

d. Cannot be determined from this data

37 [Intermediate] • Target: Applying rate-of-change reasoning to models

A function h has average rates of change of 4 on $[0, 1]$, 7 on $[1, 2]$, and 12 on $[2, 3]$. Classify the concavity of h on $[0, 3]$.

a. Linear (neither concave up nor down)

b. Cannot be determined from this data

c. Concave down

d. Concave up

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Estimating rate of change from a table • Cross-Lesson: 1.02 Rational Functions & Asymptotes

A function h has average rates of change of 9 on $[0, 1]$, 8 on $[1, 2]$, and 6 on $[2, 3]$. Classify the concavity of h on $[0, 3]$.

a. Cannot be determined from this data

b. Linear (neither concave up nor down)

c. Concave down

d. Concave up

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39 [Challenging] • Target: Visualizing rate of change on a graph • Cross-Lesson: 2.01 Exponential Functions & Growth Models

Find the average rate of change of $f(x) = 9\sqrt{x}$ on $[1, 9]$.

a. $\frac{13}{4}$

b. $\frac{9}{4}$

c. $\frac{9}{2}$

d. $\frac{1}{4}$

40 [Challenging] • Target: Concavity and real-world interpretation • Cross-Lesson: 2.02 Logarithmic Functions & Properties

Find the average rate of change of $f(x) = 5\sqrt{x}$ on $[16, 49]$.

a. $\frac{5}{11}$

b. $\frac{1}{11}$

c. $\frac{16}{11}$

d. $\frac{5}{3}$

41 [Challenging] • Target: Average Rate of Change (AROC) • Cross-Lesson: 3.01 Trigonometric Functions & the Unit Circle

Calculate the average rate of change of $f(x) = 2x^2 - 3x + 1$ on the interval $[1, 4]$.

a. 7

b. 9

c. 21

d. 6

42 [Challenging] • Target: Average Rate of Change (AROC) • Cross-Lesson: 3.02 Polar Coordinates & Polar Graphs

For a linear function $g(x) = mx + b$, what can be said about the average rate of change on any interval $[a, b]$?

a. It is constant and always equal to m

b. It depends on the midpoint $\frac{a+b}{2}$

c. It is zero if $b > 0$

d. It equals b

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43 [Challenging] • Target: Increasing/Decreasing and Changing Rates • Cross-Lesson: 4.01 Inverse & Composite Functions

If $h(x)$ is decreasing at an increasing rate on an interval, what happens to the slope of secant lines?

a. The slopes become less negative (closer to zero)

b. The slopes become more negative (steeper downward)

c. The slopes remain constant

d. The slopes switch from positive to negative

44 [Challenging] • Target: Concavity & Inflection Points • Cross-Lesson: 4.02 Piecewise-Defined & Implicit Functions

On the interval $(-\infty, 0)$, f is concave up; on $(0, \infty)$, f is concave down. If $f(0) = 4$, which statement is true?

a. $(0, 4)$ is an inflection point of f

b. $(0, 4)$ is a local minimum of f

c. $(0, 4)$ is a local maximum of f

d. $f'(0)$ must equal 0

45 [Challenging] • Target: Contextual Rates of Change Applications • Cross-Lesson: 4.03 Parametric Equations & Modeling

Water is poured into a conical cup at a constant rate. What is the concavity of the depth of water $h(t)$ as a function of time?

a. Concave down, because the radius increases with height so height increases at a decreasing rate

b. Concave up, because height increases at an increasing rate

c. Linear, because the flow rate is constant

d. Constant concavity of zero

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